

Amazon Sales Data

May 7, 2026

1 Amazon Sales Data

1.1 Description:

1.1.1 This dataset contains information on 1K+ Amazon products, including their ratings, reviews, and other details.

2 Features:

- product_id: Unique identifier for each product
- product_name: Name of the product
- category: Category of the product
- discounted_price: Discounted price of the product
- actual_price: Actual price of the product
- discount_percentage: Percentage of discount for the product
- rating: Rating of the product (1-5)
- rating_count: Number of people who voted for the Amazon rating
- about_product: Description about the product
- user_id: ID of the user who wrote the review
- user_name: Name of the user who wrote the review
- review_id: ID of the user review
- review_title: Short review
- review_content: Long review
- img_link: Image link of the product
- product_link: Official website link of the product

```
[1]: # Import some usefull libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

```
[2]: df = pd.read_csv('https://raw.githubusercontent.com/mayanktech02-lang/
↳Amazon-Sales-Analysis/refs/heads/main/amazon.csv')
```

```
[3]: df
```

```

[3]:      product_id      product_name \
0      B07JW9H4J1  Wayona Nylon Braided USB to Lightning Fast Cha...
1      B098NS6PVG  Ambrane Unbreakable 60W / 3A Fast Charging 1.5...
2      B096MSW6CT  Sounce Fast Phone Charging Cable & Data Sync U...
3      B08HDJ86NZ  boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...
4      B08CF3B7N1  Portronics Konnect L 1.2M Fast Charging 3A 8 P...
...      ...      ...
1460   B08L7J3T31  Noir Aqua - 5pcs PP Spun Filter + 1 Spanner | ...
1461   B01M6453MB  Prestige Delight PRW0 Electric Rice Cooker (1 ...
1462   B009P2LIL4  Bajaj Majesty RX10 2000 Watts Heat Convector R...
1463   B00J5DYCCA  Havells Ventil Air DSP 230mm Exhaust Fan (Pist...
1464   B01486F4G6  Borosil Jumbo 1000-Watt Grill Sandwich Maker (...

      category discounted_price \
0      Computers&Accessories|Accessories&Peripherals|...      399
1      Computers&Accessories|Accessories&Peripherals|...      199
2      Computers&Accessories|Accessories&Peripherals|...      199
3      Computers&Accessories|Accessories&Peripherals|...      329
4      Computers&Accessories|Accessories&Peripherals|...      154
...      ...      ...
1460   Home&Kitchen|Kitchen&HomeAppliances|WaterPurif...      379
1461   Home&Kitchen|Kitchen&HomeAppliances|SmallKitch...      2,280
1462   Home&Kitchen|Heating,Cooling&AirQuality|RoomHe...      2,219
1463   Home&Kitchen|Heating,Cooling&AirQuality|Fans|E...      1,399
1464   Home&Kitchen|Kitchen&HomeAppliances|SmallKitch...      2,863

      actual_price discount_percentage rating rating_count \
0           1,099           64%      4.2      24,269
1           349           43%      4.0      43,994
2           1,899           90%      3.9       7,928
3           699           53%      4.2      94,363
4           399           61%      4.2      16,905
...      ...      ...      ...      ...
1460           919           59%       4       1,090
1461          3,045           25%      4.1       4,118
1462          3,080           28%      3.6         468
1463          1,890           26%       4       8,031
1464          3,690           22%      4.3       6,987

      about_product \
0      High Compatibility : Compatible With iPhone 12...
1      Compatible with all Type C enabled devices, be...
2      Fast Charger& Data Sync -With built-in safet...
3      The boAt Deuce USB 300 2 in 1 cable is compati...
4      [CHARGE & SYNC FUNCTION]- This cable comes wit...
...      ...
1460   SUPREME QUALITY 90 GRAM 3 LAYER THIK PP SPUN F...

```

1461 230 Volts, 400 watts, 1 Year
 1462 International design and styling|Two heat sett...
 1463 Fan sweep area: 230 MM ; Noise level: (40 - 45...
 1464 Brand-Borosil, Specification â€" 23V ~ 5Hz;1 W...

user_id \
 0 AG3D604STAQKAY2UVGEUV46KN35Q,AHMY5CWJMMK5BJRBB...
 1 AECPFYFQVRUWC3KGNLJIOREFP5LQ,AGYYVPDD7YG7FYNBX...
 2 AGU3BBQ2V2DDAMOAKGFAWDDQ6QHA,AESFLDV2PT363T2AQ...
 3 AEWAZDZZJLQUYVOVGBEUKSLXHQ5A,AG5HTSFRRE6NL3M5S...
 4 AE3Q6KSUK5P75D5HFYHCRAOLODSA,AFUGIFH5ZAFXRDSZH...
 ...
 1460 AHITFY6AHALOF0HOZE0C6XBP4FEA,AFRABBODZJZQB6Z4U...
 1461 AFG5FM3NEMOL6BNFRV2NK5FNJCHQ,AGEINTRN6Z563RMLH...
 1462 AGVPWCMahyQWJQKMUJN4DW3KM5Q,AF4Q3E66MY4SR7YQZ...
 1463 AF2JQCLSCY3QJATWUNNHUSVUPNQQ,AFDMLUXC5LS5RXDJS...
 1464 AFGW5PT3R6ZAVQR4Y5MWVAKBZAYA,AG7QNJ2SCS5VS5VYY...

user_name \
 0 Manav,Adarsh gupta,Sundeep,S.Sayeed Ahmed,jasp...
 1 ArdKn,Nirbhay kumar,Sagar Viswanathan,Asp,Plac...
 2 Kunal,Himanshu,viswanath,sai niharka,saqib mal...
 3 Omkar dhale,JD,HEMALATHA,Ajwadh a.,amar singh ...
 4 rahuls6099,Swasat Borah,Ajay Wadke,Pranali,RVK...
 ...
 1460 Prabha ds,Raghuram bk,Real Deal,Amazon Custome...
 1461 Manu Bhai,Naveenpittu,Evatira Sangma,JAGANNADH...
 1462 Nehal Desai,Danish Parwez,Amazon Customer,Amaz...
 1463 Shubham Dubey,E.GURUBARAN,Mayank S.,eusuf khan...
 1464 Rajib,Ajay B,Vikas Kahol,PARDEEP,Anindya Prama...

review_id \
 0 R3HXWTOLRPONMF,R2AJM3LFTLZHFO,R6AQJGUP6P86,R1K...
 1 RGIQEG07R9HS2,R1SMWZQ86XIN8U,R2J3Y1WL29GWDE,RY...
 2 R3J3EQQ9TZI5ZJ,R3E7WBGK7IDOKV,RWU79XKQ6I1QF,R2...
 3 R3EEUZKKK9J36I,R3HJVYCLYOY554,REDECAZ7AMPQC,R1...
 4 R1BP4L2HH9TFUP,R16PVJEXKV6QZS,R2UPDB81N66T4P,R...
 ...
 1460 R3G3XFHPBFFOE8,R3COBZCD32EIGW,R2EBVBCN9QPD9R,R...
 1461 R3DDL2UPKQ2CK9,R2SYYU10ATVIU5,R1VM993161IYRW,R...
 1462 R1TLRJVV4STY5I,R2O455KRN493R1,R3Q5MVGMBRIAS2G,R...
 1463 R39Q2Y79MM9SWK,R3O79BG1NIH6MB,R29A31ZELTZNJM,R...
 1464 R2ORBRZOWEIJT9,ROKIFK9R2ISSE,R30EEG2FNJSN5I,R2...

review_title \
 0 Satisfied,Charging is really fast,Value for mo...
 1 A Good Braided Cable for Your Type C Device,Go...

2 Good speed for earlier versions,Good Product,W...
 3 Good product,Good one,Nice,Really nice product...
 4 As good as original,Decent,Good one for second...
 ...
 1460 Received the product without spanner,Excellent...
 1461 ok,everything was good couldn't return bcoz I ...
 1462 very good,Work but front melt after 2 month,Go...
 1463 Fan Speed is slow,Good quality,Good product,go...
 1464 Works perfect,Ok good product,Nice Product. Re...

review_content \

0 Looks durable Charging is fine tooNo complains...
 1 I ordered this cable to connect my phone to An...
 2 Not quite durable and sturdy,https://m.media-a...
 3 Good product,long wire,Charges good,Nice,I bou...
 4 Bought this instead of original apple, does th...
 ...
 1460 I received product without spanner,Excellent p...
 1461 ok,got everything as mentioned but the measuri...
 1462 plastic but cool body ,u have to find sturdy s...
 1463 I have installed this in my kitchen working fi...
 1464 It does it job perfectly..only issue is temp c...

img_link \

0 https://m.media-amazon.com/images/W/WEBP_40237...
 1 https://m.media-amazon.com/images/W/WEBP_40237...
 2 https://m.media-amazon.com/images/W/WEBP_40237...
 3 https://m.media-amazon.com/images/I/41V5FtEWPk...
 4 https://m.media-amazon.com/images/W/WEBP_40237...
 ...
 1460 https://m.media-amazon.com/images/I/41fDdRtjfx...
 1461 https://m.media-amazon.com/images/I/41gzDxk4+k...
 1462 https://m.media-amazon.com/images/W/WEBP_40237...
 1463 https://m.media-amazon.com/images/W/WEBP_40237...
 1464 https://m.media-amazon.com/images/W/WEBP_40237...

product_link

0 https://www.amazon.in/Wayona-Braided-WN3LG1-Sy...
 1 https://www.amazon.in/Ambrane-Unbreakable-Char...
 2 https://www.amazon.in/Sounce-iPhone-Charging-C...
 3 https://www.amazon.in/Deuce-300-Resistant-Tang...
 4 https://www.amazon.in/Portronics-Konnect-POR-1...
 ...
 1460 https://www.amazon.in/Noir-Aqua-Spanner-Purifi...
 1461 https://www.amazon.in/Prestige-Delight-PRW0-1-...
 1462 https://www.amazon.in/Bajaj-RX-10-2000-Watt-Co...
 1463 https://www.amazon.in/Havells-Ventilair-230mm-...

1464 <https://www.amazon.in/Borosil-Jumbo-1000-Watt-...>

[1465 rows x 16 columns]

```
[4]: df.isnull().sum() # check null value in data
```

```
[4]: product_id          0
     product_name       0
     category          0
     discounted_price   0
     actual_price       0
     discount_percentage 0
     rating            0
     rating_count       2
     about_product      0
     user_id           0
     user_name          0
     review_id         0
     review_title       0
     review_content     0
     img_link          0
     product_link       0
     dtype: int64
```

```
[6]: df.dropna(inplace = True) # drop the null values, because only 2 palce found in
     ↪rating_count column where null value present.
```

```
[9]: df.duplicated().sum() #check duplicate value in data, in that dataset no
     ↪duplicate data found
```

```
[9]: np.int64(0)
```

```
[48]: df.info()
```

```
<class 'pandas.DataFrame'>
Index: 1462 entries, 0 to 1464
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype
---  -
0   product_id            1462 non-null   str
1   product_name          1462 non-null   str
2   category              1462 non-null   str
3   discounted_price      1462 non-null   float64
4   actual_price          1462 non-null   float64
5   discount_percentage   1462 non-null   int64
6   rating               1462 non-null   float64
7   rating_count          1462 non-null   str
8   about_product        1462 non-null   str
```

```

9   user_id          1462 non-null   str
10  user_name        1462 non-null   str
11  review_id        1462 non-null   str
12  review_title     1462 non-null   str
13  review_content   1462 non-null   str
14  img_link         1462 non-null   str
15  product_link     1462 non-null   str
dtypes: float64(3), int64(1), str(12)
memory usage: 194.2 KB

```

```

[20]: df.discounted_price = df.discounted_price.str.replace(' ', '') #remove the rupee
      ↪sign from value
df.discounted_price = df.discounted_price.str.replace(',', '')

```

```

[27]: df.discounted_price = df.discounted_price.astype(float)

```

```

[24]: df.actual_price = df.actual_price.str.replace(' ', '')
df.actual_price = df.actual_price.str.replace(',', '')

```

```

[29]: df.actual_price = df.actual_price.astype(float)

```

```

[31]: df.discount_percentage = df.discount_percentage.str.replace('%', '')
df.discount_percentage = df.discount_percentage.str.replace(',', '')

```

```

[32]: df.discount_percentage = df.discount_percentage.astype(int)

```

```

[45]: df = df[df["rating"] != ''] # check the data where the string value

```

```

[35]: df.rating = df.rating.str.replace('|', '')

```

```

[47]: df.rating = df.rating.astype(float)

```

```

[50]: df.rating_count = df.rating_count.str.replace(',', '')

```

```

[51]: df.rating_count = df.rating_count.astype(float)

```

```

[102]: df.info()

```

```

<class 'pandas.DataFrame'>
Index: 1462 entries, 0 to 1464
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   product_id            1462 non-null   str
1   product_name          1462 non-null   str
2   category              1462 non-null   str
3   discounted_price      1462 non-null   float64
4   actual_price          1462 non-null   float64
5   discount_percentage   1462 non-null   int64

```

```

6   rating                1462 non-null   float64
7   rating_count          1462 non-null   float64
8   about_product         1462 non-null   str
9   user_id               1462 non-null   str
10  user_name             1462 non-null   str
11  review_id             1462 non-null   str
12  review_title          1462 non-null   str
13  review_content        1462 non-null   str
14  img_link              1462 non-null   str
15  product_link          1462 non-null   str
16  category_1            1462 non-null   str
17  category_2            1462 non-null   str
18  category_3            1454 non-null   str
19  category_4            1297 non-null   str
dtypes: float64(4), int64(1), str(15)
memory usage: 239.9 KB

```

```

[103]: cat_cols = df['category'].str.split('|', expand=True)

print(cat_cols.head())

```

	0	1	2	3 \
0	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	Cables
1	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	Cables
2	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	Cables
3	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	Cables
4	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	Cables

	4	5	6
0	USBCables	NaN	NaN
1	USBCables	NaN	NaN
2	USBCables	NaN	NaN
3	USBCables	NaN	NaN
4	USBCables	NaN	NaN

```

[66]: cat_cols.columns = [f'category_{i+1}' for i in range(cat_cols.shape[1])]

```

```

[126]: df.drop(['category'], axis = 1, inplace= True)

```

```

[100]: df = pd.concat([df, cat_cols], axis=1)

```

```

[132]: pd.reset_option('all')

```

3 1 What is the average rating for each product categoryk

```

[140]: df.groupby('category')['rating'].mean().sort_values(ascending=False)

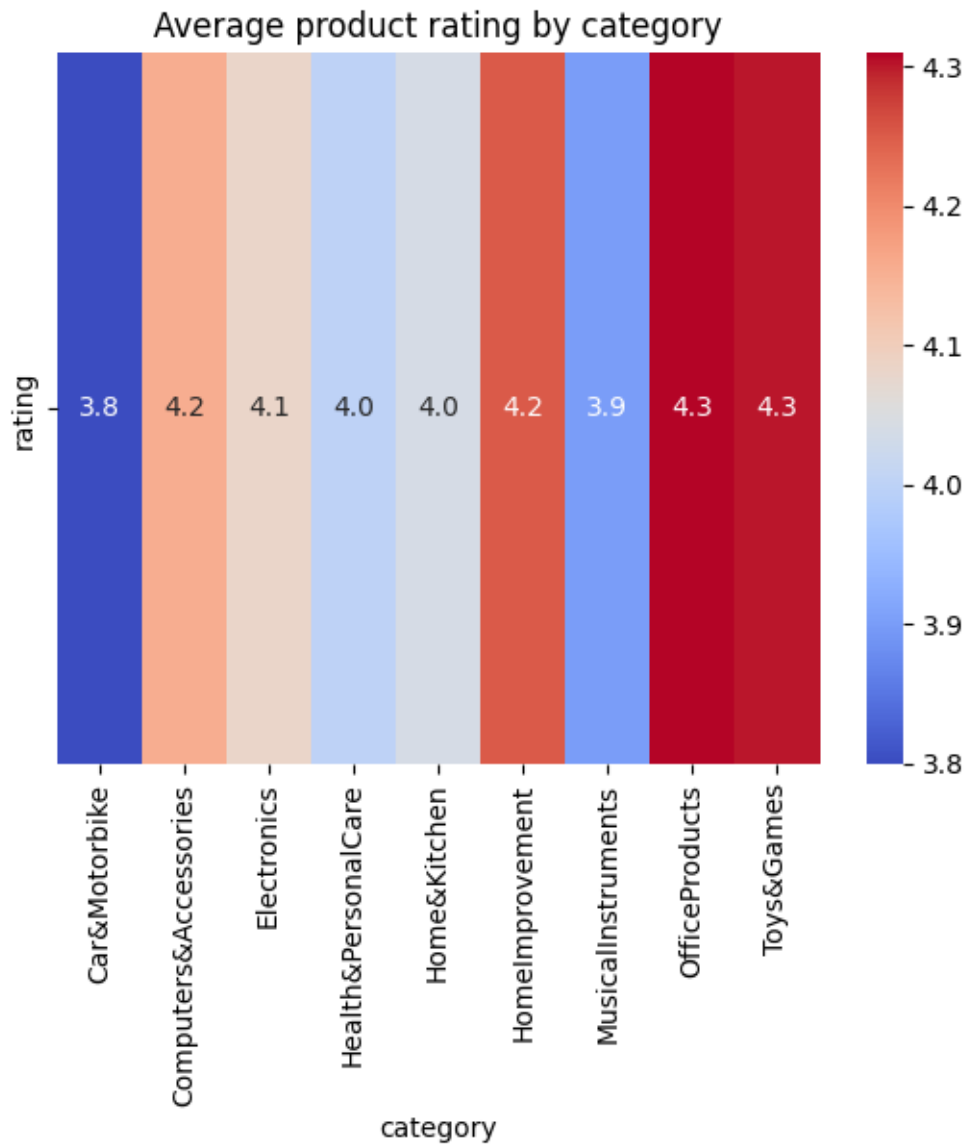
```

```
[140]: category
      OfficeProducts      4.309677
      Toys&Games         4.300000
      HomeImprovement    4.250000
      Computers&Accessories 4.155654
      Electronics        4.081749
      Home&Kitchen        4.040716
      Health&PersonalCare 4.000000
      MusicalInstruments  3.900000
      Car&Motorbike       3.800000
      Name: rating, dtype: float64
```

```
[142]: pivot_table = df.pivot_table(
      values='rating',
      columns='category',
      aggfunc='mean'
    )
```

```
[146]: plt.figure()
      sns.heatmap(
        pivot_table,
        annot=True,
        fmt='.1f',
        cmap='coolwarm'
      )

      plt.title('Average product rating by category')
      plt.show()
```



4 2. What are the top rating_count products by category

```
[164]: top_products = df.loc[
        df.groupby('category')['rating_count'].idxmax(),
        ['category', 'product_name', 'rating_count']
    ]
```

```
[165]: top_products.sort_values(by = 'rating_count', ascending=False)
```

```
[165]:          category \
12          Electronics
1028         Home&Kitchen
588  Computers&Accessories
613    MusicalInstruments
924         Toys&Games
659         OfficeProducts
982         HomeImprovement
1247    Health&PersonalCare
1157         Car&Motorbike

          product_name  rating_count
12    AmazonBasics Flexible Premium HDMI Cable (Blac...    426973.0
1028  Pigeon Polypropylene Mini Handy and Compact Ch...    270563.0
588          SanDisk Cruzer Blade 32GB USB Flash Drive    253105.0
613  Boya ByM1 Auxiliary Omnidirectional Lavalier C...    68664.0
924  Faber-Castell Connector Pen Set - Pack of 25 (...    15867.0
659  Casio FX-82MS 2nd Gen Non-Programmable Scienti...    12179.0
982  Gizga Essentials Cable Organiser, Cord Managem...    5985.0
1247  Dr Trust Electronic Kitchen Digital Scale Weig...    3663.0
1157  Reffair AX30 [MAX] Portable Air Purifier for C...    1118.0
```

5 3. What is the distribution of discounted prices vs. actual prices

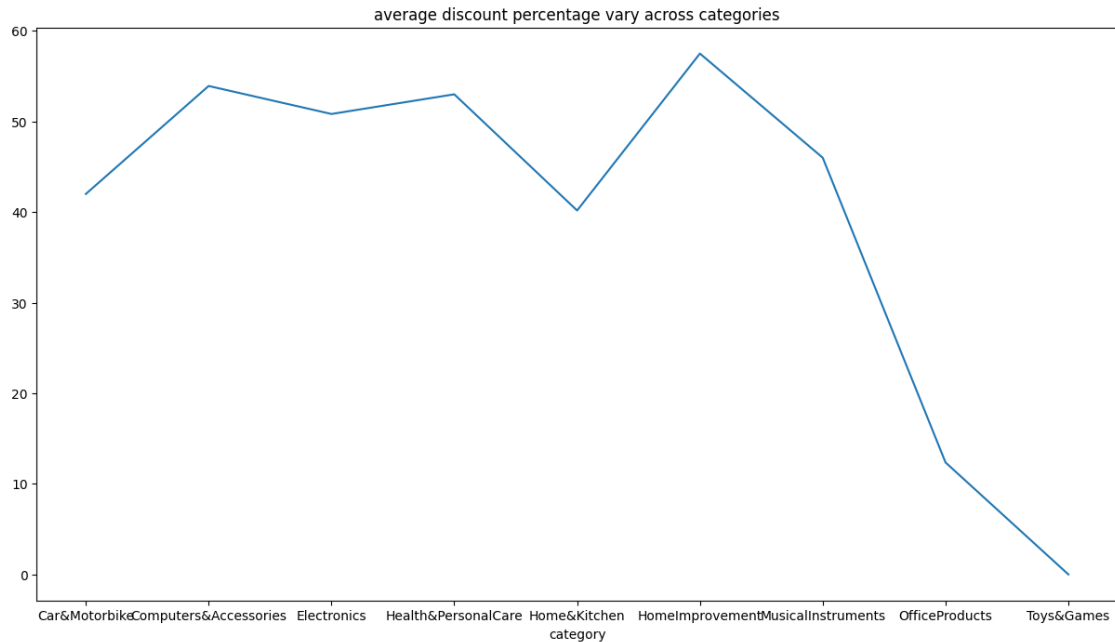
```
sns.scatterplot(df, x='actual_price', y='discounted_price')
```

```
[168]: df.columns
```

```
[168]: Index(['product_id', 'product_name', 'discounted_price', 'actual_price',
        'discount_percentage', 'rating', 'rating_count', 'about_product',
        'user_id', 'user_name', 'review_id', 'review_title', 'review_content',
        'img_link', 'product_link', 'category'],
        dtype='str')
```

6 4. How does the average discount percentage vary across categories

```
[183]: plt.figure(figsize=(15,8))
df.groupby('category')['discount_percentage'].mean().plot()
plt.title('average discount percentage vary across categories')
plt.show()
```



7 5. What are the most popular product names

```
[185]: popular_products = df[['product_name', 'rating_count']].sort_values(
        by='rating_count',
        ascending=False
    )

popular_products.head(10)
```

```
[185]:
```

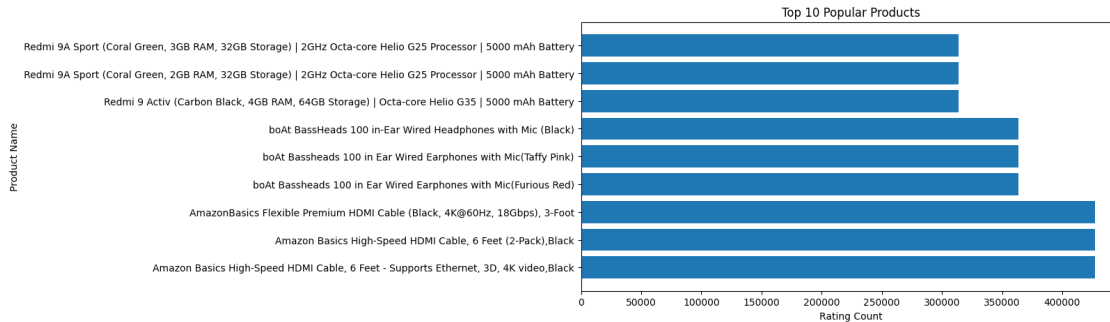
	product_name	rating_count
47	Amazon Basics High-Speed HDMI Cable, 6 Feet - ...	426973.0
65	Amazon Basics High-Speed HDMI Cable, 6 Feet (2...	426973.0
12	AmazonBasics Flexible Premium HDMI Cable (Blac...	426973.0
684	AmazonBasics Flexible Premium HDMI Cable (Blac...	426972.0
400	boAt Bassheads 100 in Ear Wired Earphones with...	363713.0
352	boAt Bassheads 100 in Ear Wired Earphones with...	363713.0
584	boAt BassHeads 100 in-Ear Wired Headphones wit...	363711.0
370	Redmi 9 Activ (Carbon Black, 4GB RAM, 64GB Sto...	313836.0
371	Redmi 9A Sport (Coral Green, 2GB RAM, 32GB Sto...	313836.0
566	Redmi 9A Sport (Coral Green, 3GB RAM, 32GB Sto...	313832.0

```
[187]: top10 = popular_products.head(10)

plt.figure(figsize=(10,5))
plt.barh(top10['product_name'], top10['rating_count'])
```

```
plt.xlabel('Rating Count')
plt.ylabel('Product Name')
plt.title('Top 10 Popular Products')

plt.show()
```



8 5. What are the most popular product keywords

```
[188]: import re
from collections import Counter

text = ' '.join(df['product_name'].astype(str))

# keep only letters and numbers
text = re.sub(r'[^a-zA-Z0-9 ]', '', text)

words = text.lower().split()

keyword_count = Counter(words)

print(keyword_count.most_common(20))
```

```
[('with', 749), ('for', 672), ('black', 466), ('cable', 399), ('usb', 390),
('and', 329), ('charging', 244), ('to', 214), ('fast', 208), ('c', 185),
('smart', 172), ('white', 154), ('type', 147), ('tv', 145), ('compatible', 141),
('data', 134), ('1', 134), ('hd', 129), ('3', 118), ('in', 118)]
```

9 6. What are the most popular product reviews

```
[192]: top_reviews = df[['product_name', 'rating', 'rating_count', 'review_title',
    ↪ 'review_content']].sort_values(
    by='rating_count',
```

```

        ascending=False
    )

top_reviews.head(10)

```

```

[192]:

```

	product_name	rating	rating_count	\
47	Amazon Basics High-Speed HDMI Cable, 6 Feet - ...	4.4	426973.0	
65	Amazon Basics High-Speed HDMI Cable, 6 Feet (2...	4.4	426973.0	
12	AmazonBasics Flexible Premium HDMI Cable (Blac...	4.4	426973.0	
684	AmazonBasics Flexible Premium HDMI Cable (Blac...	4.4	426972.0	
400	boAt Bassheads 100 in Ear Wired Earphones with...	4.1	363713.0	
352	boAt Bassheads 100 in Ear Wired Earphones with...	4.1	363713.0	
584	boAt BassHeads 100 in-Ear Wired Headphones wit...	4.1	363711.0	
370	Redmi 9 Activ (Carbon Black, 4GB RAM, 64GB Sto...	4.1	313836.0	
371	Redmi 9A Sport (Coral Green, 2GB RAM, 32GB Sto...	4.1	313836.0	
566	Redmi 9A Sport (Coral Green, 3GB RAM, 32GB Sto...	4.1	313832.0	

	review_title	\
47	It's quite good and value for money,Works well...	
65	It's quite good and value for money,Works well...	
12	It's quite good and value for money,Works well...	
684	It's quite good and value for money,Works well...	
400	Best value for money,HEAD PHONE POUCH NOT RECE...	
352	Best value for money,HEAD PHONE POUCH NOT RECE...	
584	Best value for money,HEAD PHONE POUCH NOT RECE...	
370	Best phone for below normal use,Good mobile fo...	
371	Best phone for below normal use,Good mobile fo...	
566	Best phone for below normal use,Good mobile fo...	

	review_content
47	I am using it for 14 days now. The experience ...
65	I am using it for 14 days now. The experience ...
12	I am using it for 14 days now. The experience ...
684	I am using it for 14 days now. The experience ...
400	The sound quality of this earphone are really ...
352	The sound quality of this earphone are really ...
584	The sound quality of this earphone are really ...
370	If you want a smart phone for just the use of ...
371	If you want a smart phone for just the use of ...
566	If you want a smart phone for just the use of ...

```

[194]: df.columns

```

```

[194]: Index(['product_id', 'product_name', 'discounted_price', 'actual_price',
        'discount_percentage', 'rating', 'rating_count', 'about_product',
        'user_id', 'user_name', 'review_id', 'review_title', 'review_content',
        'img_link', 'product_link', 'category'],

```

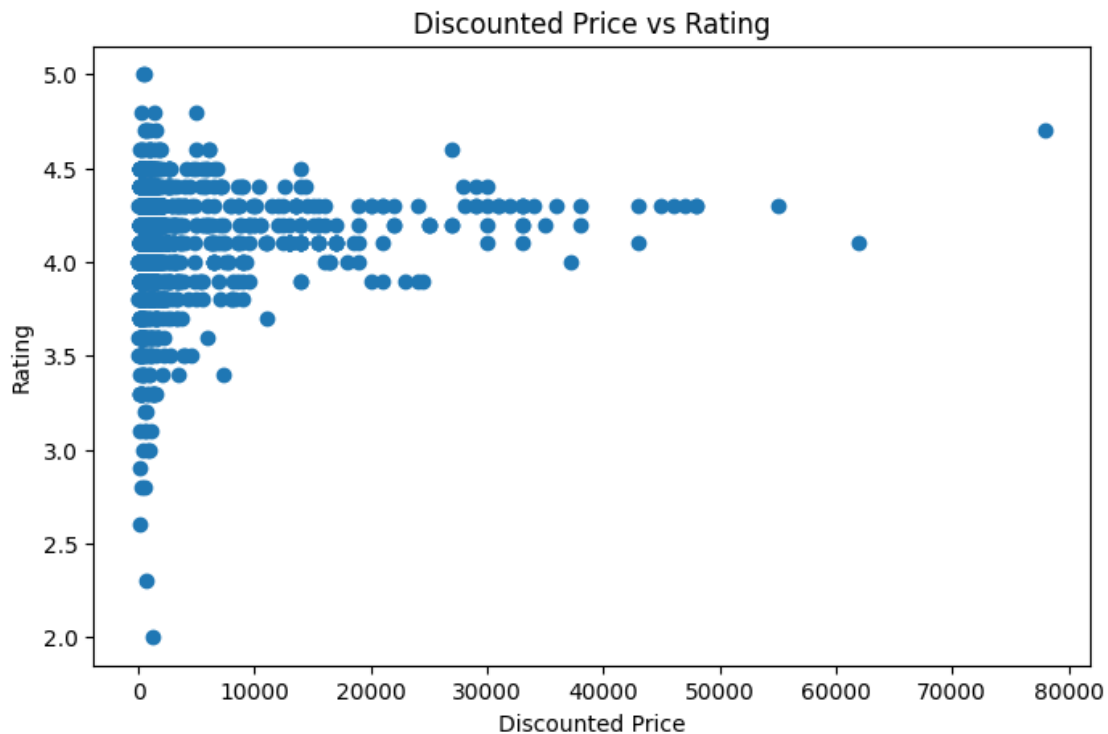
```
dtype='str')
```

10 7. What is the correlation between discounted_price and rating

```
[196]: correlation = df['discounted_price'].corr(df['rating'])  
  
print(correlation)
```

```
0.12113187526066276
```

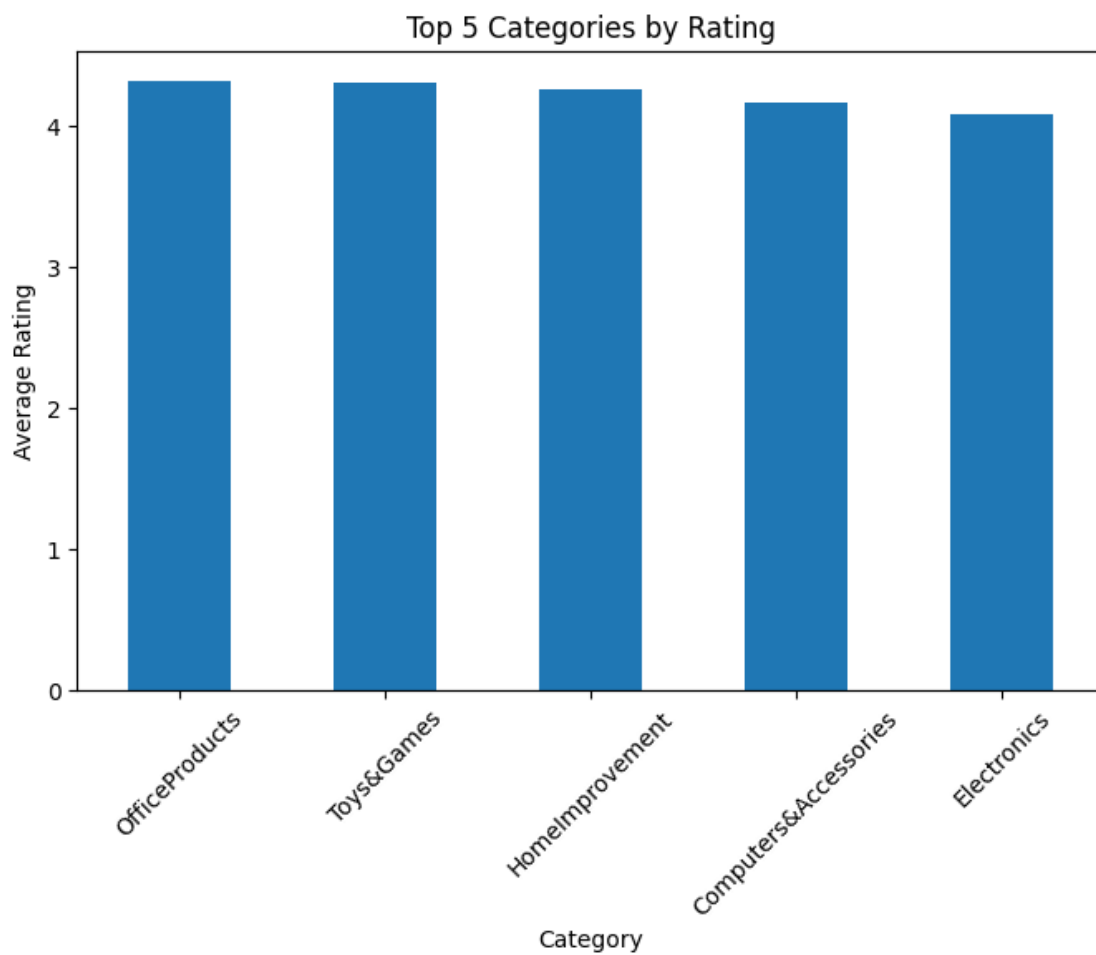
```
[197]: plt.figure(figsize=(8,5))  
  
plt.scatter(df['discounted_price'], df['rating'])  
  
plt.xlabel('Discounted Price')  
plt.ylabel('Rating')  
plt.title('Discounted Price vs Rating')  
  
plt.show()
```



11 9. What are the Top 5 categories based on the highest ratings

```
[211]: highest_rating = df.groupby('category')['rating'].mean().round(2).  
       ↪sort_values(ascending=False).head(5)
```

```
[213]: highest_rating.plot(  
        kind='bar',  
        figsize=(8,5)  
    )  
  
plt.xlabel('Category')  
plt.ylabel('Average Rating')  
plt.title('Top 5 Categories by Rating')  
  
plt.xticks(rotation=45)  
  
plt.show()
```



12 10 Identify any potential areas for improvement or optimization based on the data analysis.

Low-rated categories Improve quality & customer experience

Low-review products Increase marketing visibility

Poor high-volume products Fix common customer complaints

Keyword optimization Improve SEO & search ranking

Pricing issues Align price with customer expectations Discount strategy Focus on value, not only discounts

[]:

[]: